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Advantages of developing logical DFD

- ① Represent how the business operate and include the related process of registration.
 - ② Better communication with users. They are familiar with how the business operates how the
 - ③ Increased flexibility & easy to maintain
The system will have
- Clarifying which process are performed by humans (manual) and which are automated. These control measurements to ensure accuracy and completeness.
 - describing process in more detail
 - which depicts more processes with specific step-by-step automated process in sequence order to be executed.
 - identifying temporary data stores and transaction file
 - storing specific records of related data with specific and different record names.

Goals for producing mini specification :

- 1) Reduce process ambiguity
- 2) Obtain a precise description of what is accomplished
- 3) Validate the system design, including the data flow diagrams and data dictionary

NOT created for :

- 1) Processes that represent physical input / output.
- 2) " simple data validation
- 3) " use prewritten code

Structured English

START

DO WHILE

GET

ENDDO

BEGIN IF

IF XXX Then

—

—

IF — Then

—

—

ELSE

—

ENDIF

ELSE

END IF

END

Decision Tree

- complex branching occurs in a structured decision process.

Advantages:

- 1) Condition and action of decision are found on the same branches but not on others.
- 2) Compare to decision table, more readily understood by others in the organization.

Decision Table

Conditions and Actions	Rules
Conditions	Condition Alternatives
Actions	Action Entries

Advantages:

- 1) help the analysis ensure completeness
- 2) easy to check for possible error
 - Impossible situations
 - Redundancy
 - Contradictions

When to Use ?

Advantages

Structured English

when they are many repetitions action or when communication to end users is important.

Decision table

when complex combination of condition, action and rules is found, or require a method to avoid impossible situation, contradiction, redundancy.

Decision Tree

when the sequence of conditions and action is important, or when not every conditions is relevant to each actions.

} describe complex decision logic

Chapter 5

Output Design Objective

-designing the output to serve the intended purpose

Designing Printed Output Reports

1) Detail reports

- print a report line for every record on the master file
- quite lengthy

2) Exception reports

- print a line for all records that match a certain condition
- Useful when the user wants specific information

3) Summary reports

- print one line for a group report that are used to make decisions
 - Reports that provide comprehensive data.
- used by individuals at higher level in the organization

Input Method

① Batch input - data entry performed on a specific time schedule, such as daily, weekly, monthly or longer

↳ -offline, processing quickly, non peak times

↳ centralized activity, processing is delay

② Online data entry - Sources data automation

↳ immediately feedback, update

↳ costly, data entry procedure may be lacking

Input form design guideline

- 1) Make forms easy to fill out (proper flow, grp into logically, provides clear caption)
- 2) meet the purpose
- 3) assure accurate completion (reduce input details) reduce error rate
- 4) keep forms attractive (different fonts in the same form)
uncluttered

Good display Screen Guidelines

1) Keep display screen simple

- 3 sections :
- Heading
 - Body
 - Comments and instructions

2) keep the display presentation consistent

3) Facilitate user movement among display screens.

4) attractive display screen and pages.

- Graphical User interfaces (GUI) are used with a mouse for entering data & making selections.

- Icon
 - run programs
 - execute command

- ① Effective - supports business requirements
- ② Reliable - processing errors, handle input error
- ③ Maintainable - flexible

UI Design Guidelines

- ① Focus on system design objective → improve user efficiency and productivity
help users to perform their jobs
- ② Create an interface that is easy to learn and use
 - clearly label all controls, button, and icons
 - select only those images that a user can understand easily.
 - show all commands in a list of menu items
 - make it easy to navigate or return to any level in menu structure
- ③ Provide features that promote efficiency
 - create alphabetical menu lists
 - create shortcut
 - Use default values if the majority of values in a field are the same.
- ④ Make it easy for users.
 - ensure that help is always available
- ⑤ Minimize input data problem
 - Set input format : like MM DD YY
 - Use drop down menu
 - Use check box
 - establish a list of predefined values that user can click to select.

⑥ Provide feedback to the users

⑦ Create attractive layout and design

- use appropriate colours
- group related objects & info
- use hyperlinks to jump to related topics

⑧ Use familiar terms and images

- use familiar commands

System implementation

Purpose / importance :

- 1) To convert final physical system specifications into working and reliable software.
- 2) To document work that has been done
- 3) To provide help for current and future users.

System Construction (coding) & Testing

development of all parts of the system
(Programming)

a process of analyzing a software item to detect the differences between existing and required conditions (detect)

Acceptance Test

System Test

Integration Test

Unit Test

• User Acceptance Test

• Operational acceptance Testing

• Alpha Test

• Regulation acceptance Testing

• Beta Test

performed at developer's site by customer

performed at the customer site by end user / customer

Installation $\Rightarrow$ the organizational process of changing over from current information system to a new one.

Software installation

hardware installation

data conversion

X ensure the phased transition between the old and new program

Direct Installation - turning off the old system, new system is turned on

✓ No transition costs

X High Risk

No duplication of effort by the user

Parallel Installation - running the old information and the new one at the same time

✓ Low Risk

X Transition cost

Duplication of effort by the user

Single location installation - trying out an information system at one site first
use the experience to decide if and how the new system will
be deployed

✓ Low Risk at different location

Phased Installation - changing from the old system to a new one incrementally

- starting with one or few functional components and then gradually extending the installation to the whole system

✓ it give user learning advantage

✓ Low Risk where there is possibility of detecting and recovering from errors without a lot of down time,

✓ Easier installation - incremental

X Long period of change

Planning Installation - Consideration

Documentation

① System documentation - detail information

② User documentation - how it works and how to use it

③ Internal documentation - program source code

④ External documentation - data flow and E-R diagrams

Training and supporting users



- Use of the system
- system installation
- system management

provide ongoing educational and
problem-solving assistance

Training Methods

- ① Resident expert
- ② E-learning, distance learning
- ③ Blended learning (instructor plus e-learning)
- ④ Software help components
- ⑤ External sources (Vendors)
- ⑥ Traditional instructor-led classroom training

Organizational issues

security issues
project close-down

- management support
- user involvement
- commitment to change
- to project
- Extend of project definition and planning.

Use Case Diagrams

Actor Primary 在左, secondary 在右

Someone or something that uses our system to achieve a goal.

- Person
- Organization
- Another system
- An external device

Use case symbol - start with verb
 - put in logical order

★ each actor has to interact at least one use cases.

Includes Relationship

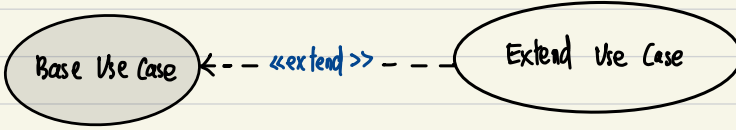
★ happens everytime



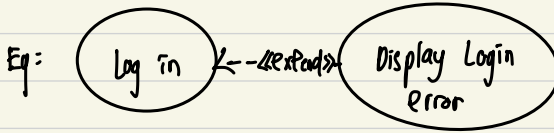
★ 同时进行

★ log in use case won't be complete unless verify password is complete.

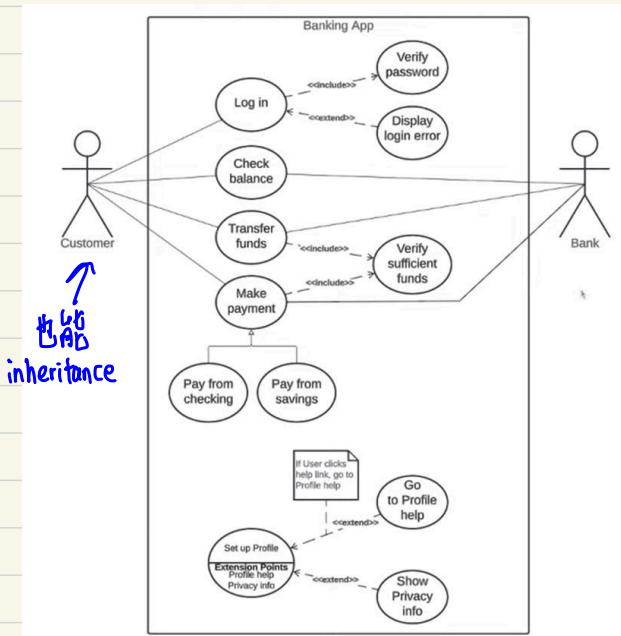
Extend Relationship



★ When base use case executed, extend use case will happen sometime, but not everytime.



Generalization Relationship (Inheritance)



Activity Diagram

→ a flowchart to represent the flow of control among the activities in a system

→ The flow of operation can be sequential, branched or concurrent.

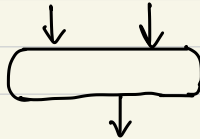
1. Start Node ●

2. Final Activity Node ⊙

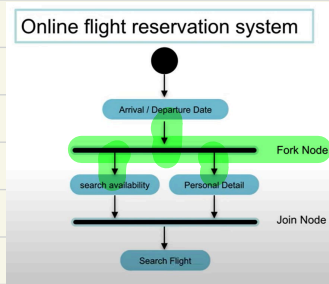
3. Activity Activity

4. Control Flow →

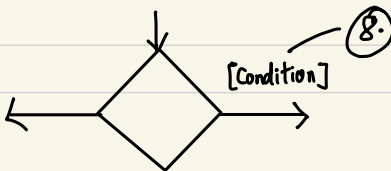
5. Join Node



6. Fork Node

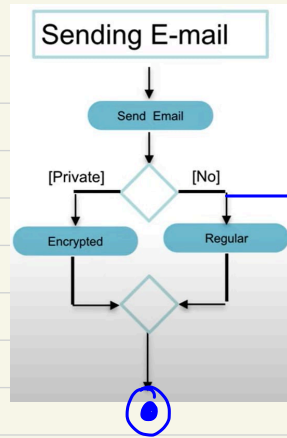
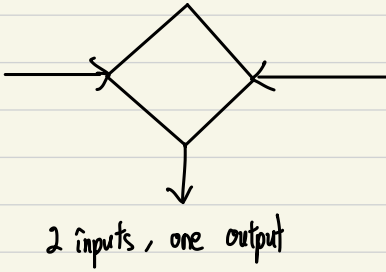


7. Decision



2 outputs, 1 input

9. Merge Node

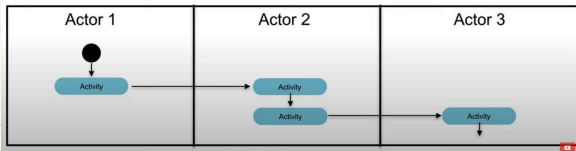


10. Final Flow Node



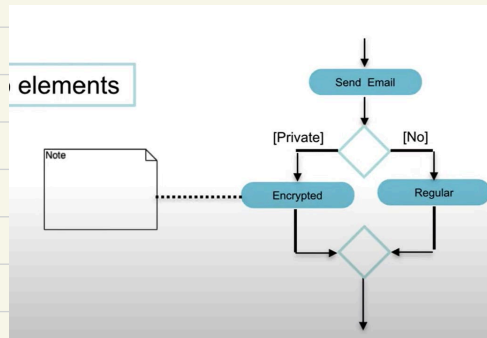
11. Partition

Also known as **swim lanes** which are used to represent or group actions carried out by different actors in a single thread



12. Note

- Use to add relevant comments to elements



13. Signal Sending

Signal Sending

14. Signal Receipt

Signal Receipt

